

eFS API Deployment and ORDS–OAuth Configuration Guide

This guide provides detailed steps to enable, configure, and test Oracle REST Data Services (ORDS) with OAuth for eFS API deployment.

0. Enable and Register ORDS Schema

Ensure ORDS is enabled for the schema and the REST API source script (RestAPI.sql) has been successfully deployed.

1. Update Global Rule Configuration

Update application server details in EMOB.MB_GLOBAL_RULE:

```
UPDATE EMOB.MB_GLOBAL_RULE SET APP_SERVER_IP = '172.16.11.3', APP_SERVER_PORT = '8080',  
APP_SERVER_PATH = '/ufl/efs/';
```

2. Create ORDS Privilege and Role

Create Role and Privilege

```
DECLARE la_roles OWA.vc_arr; la_priv_patterns OWA.vc_arr; BEGIN ords.create_role (p_role_name  
=> 'ERA_USER_ROLE'); la_roles (1) := 'ERA_USER_ROLE'; la_priv_patterns (1) :=  
'/customers_loan/tax_statement'; ords.define_privilege ( p_privilege_name =>  
'API_DATA_ACCESS', p_roles => la_roles, p_patterns => la_priv_patterns, p_label => 'ERA Data  
Access', p_description => 'Provides the ability to access eFS Data.' ); COMMIT; END;
```

Verify Role and Privilege:

```
SELECT * FROM user_ords_roles WHERE name = 'ERA_USER_ROLE'; SELECT * FROM  
user_ords_privileges WHERE name = 'API_DATA_ACCESS';
```

3. Delete Privilege (If Needed)

```
BEGIN ords.delete_privilege (p_name => 'API_DATA_ACCESS'); COMMIT; END;
```

4. Create OAuth Client and Grant Role

```
BEGIN OAuth.create_client ( p_name => 'TREASURY_APP', p_grant_type => 'client_credentials',  
p_description => 'ERA User', p_support_email => 'info@erainfotechbd.com', p_privilege_names  
=> NULL ); OAuth.grant_client_role ( p_client_name => 'TREASURY_APP', p_role_name =>  
'ERA_USER_ROLE' ); COMMIT; END;
```

Verify OAuth Clients:

```
SELECT x.* FROM user_ords_clients x;
```

5. Access Token Information

Access Token = T-PZPjySsQiwKKc9V37PJA Header Prefix = Bearer

6. Modify Token Expiry Time

By default, ORDS OAuth tokens expire after 3600 seconds (1 hour). To increase token lifetime to 7200 seconds (2 hours): 1. Open the ORDS configuration file: \$ORDS_CONF/global/defaults.xml 2. Add or update: 7200 3. Restart ORDS (standalone) or reload (Tomcat/WebLogic/GlassFish).

7. Change Client ID or Secret

```
UPDATE user_ords_clients SET CLIENT_ID = 'ERA_TREASURY_APP', CLIENT_SECRET =  
'ERA_TRS_UFIL_LIVE' WHERE NAME LIKE 'TREASURY_APP';
```

8. Disable SSL Verification (Testing Only)

If Postman or any API client fails to generate a token due to SSL verification errors, add this to ORDS config. Older ORDS: false Latest ORDS: Add inside /u02/ords_conf/global/settings.xml Then restart Tomcat or application server.

Test Token Retrieval in Postman:

Endpoint: http://172.16.11.3:8080/ufl/efs/cbs/get/token (GET) Headers: pClientId = ERA_TREASURY_APP pClientSecret = ERA_TRS_UFIL_LIVE

9. Testing API from SQL or TOAD

SELECT UTL_HTTP.REQUEST(:A) AS RESPONSE FROM DUAL; Replace :A with your API endpoint URL.

10. Summary of Key Objects

Type	Name	Description
ORDS Role	ERA_USER_ROLE	Grants access to API endpoints
Privilege	API_DATA_ACCESS	/customers_loan/tax_statement
OAuth Client	TREASURY_APP	Connected app using client_credentials
Token Lifetime	7200 sec (2 hours)	Configurable in defaults.xml
SSL Verify	false (optional)	Only for testing in Postman

11. Deployment Recommendations

- Deploy and test in a staging environment first. - Backup configuration files (defaults.xml, settings.xml). - Keep Client Secret confidential. - Restart ORDS or Tomcat after configuration changes. - Use Postman environment variables for credentials and tokens. - Verify access with both Postman and UTL_HTTP.